Operations Analytics

Service requests, SLA pressure and the delays hiding between teams

Decision brief | Sample period July 2025 to mid-January 2026 | Snapshot date 2026-01-15

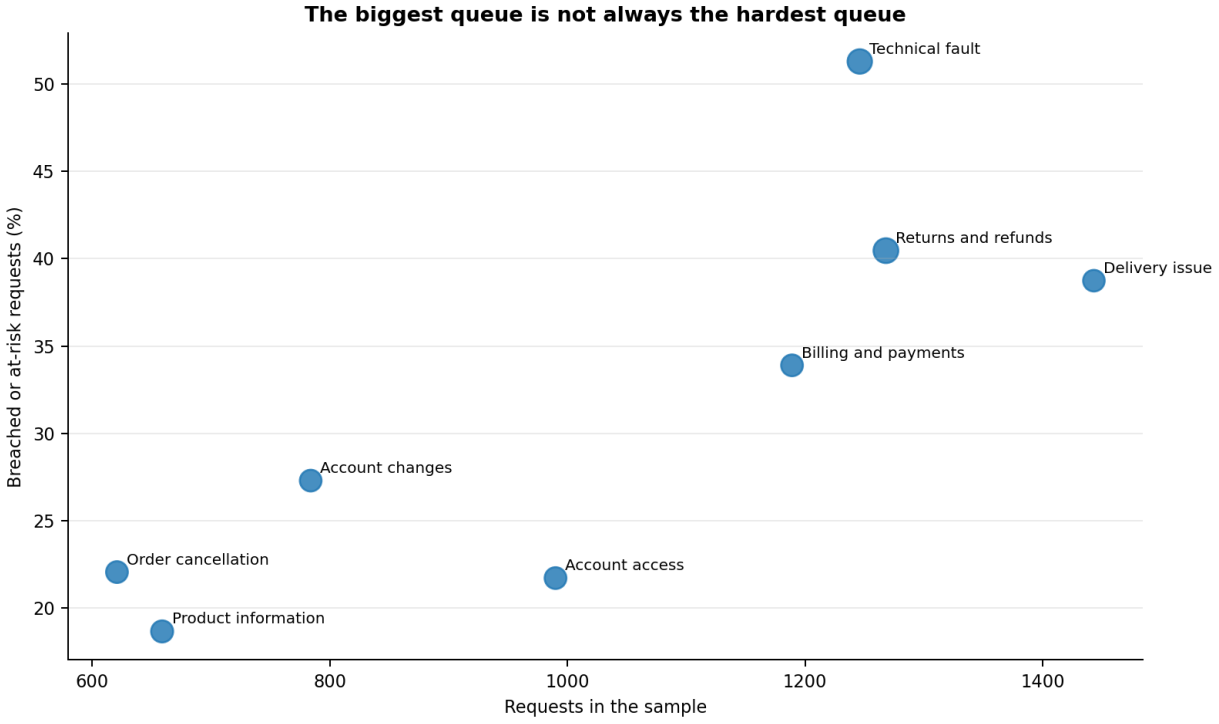
Requests	Resolved	Still open	SLA met	Median (h)	P90 (h)
8,200	8,105	95	66.5%	13.8 h	47.1 h

What the numbers are saying

There are 8,200 requests in this sample, of which 95 are still open at the snapshot date. Among resolved requests, 66.5% finished within their target. The median resolution time is 13.8 hours, but the 90th percentile is 47.1 hours. That gap is the operational story: the typical ticket and the ticket customers remember are often not the same ticket.

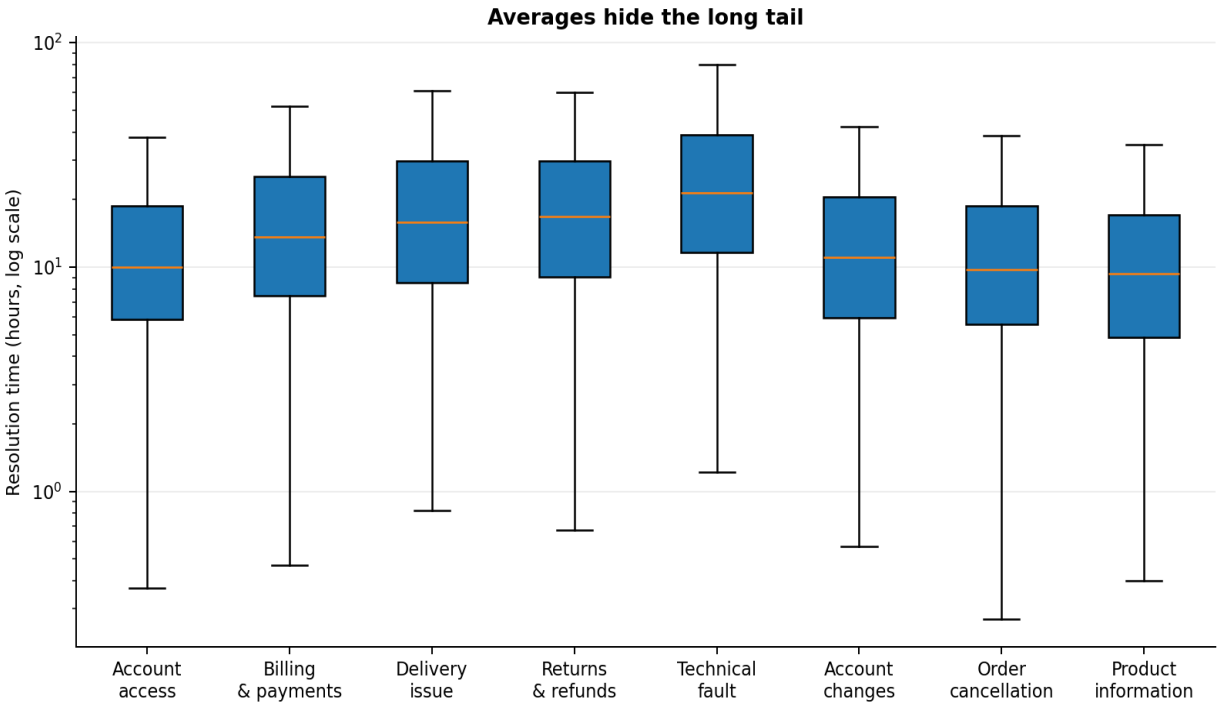
89 open requests have already used more than their resolution target. That is a more actionable watchlist than a single backlog count, because it separates unfinished work that is still within its promise from work that has already crossed it.

Read the shape of the queue

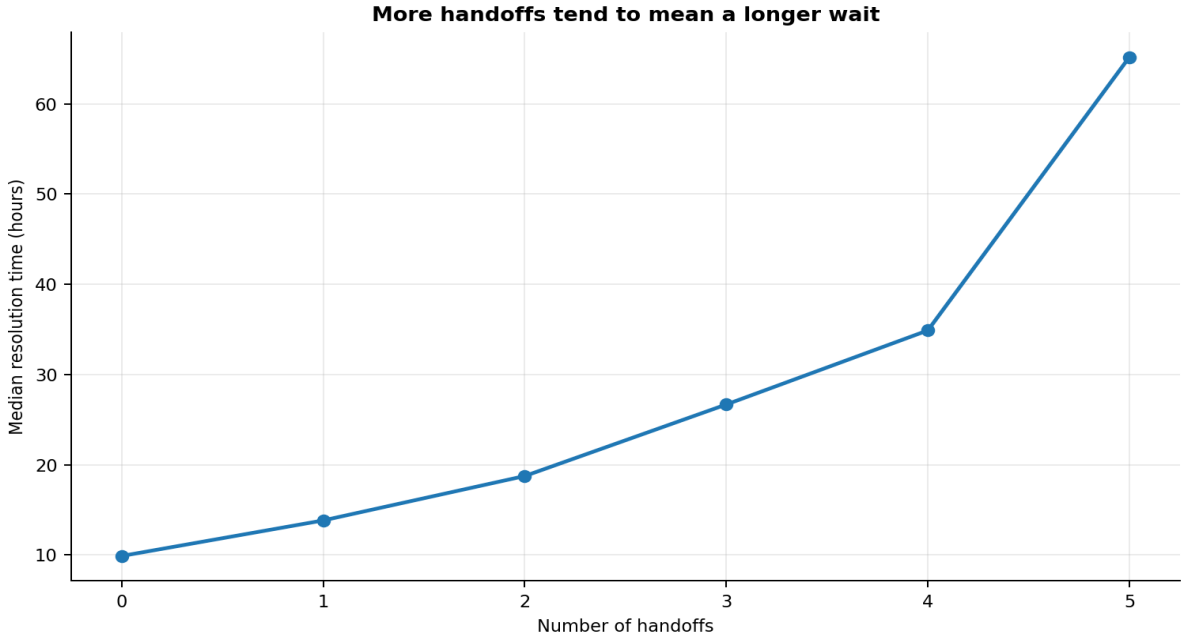


This view puts volume beside the share of requests that have breached or are at risk of breaching their target. Bubble size indicates average handoffs. The work to tackle first is not automatically the biggest category; look for queues that combine meaningful volume, high risk and repeated movement between teams.

The long tail is the service problem

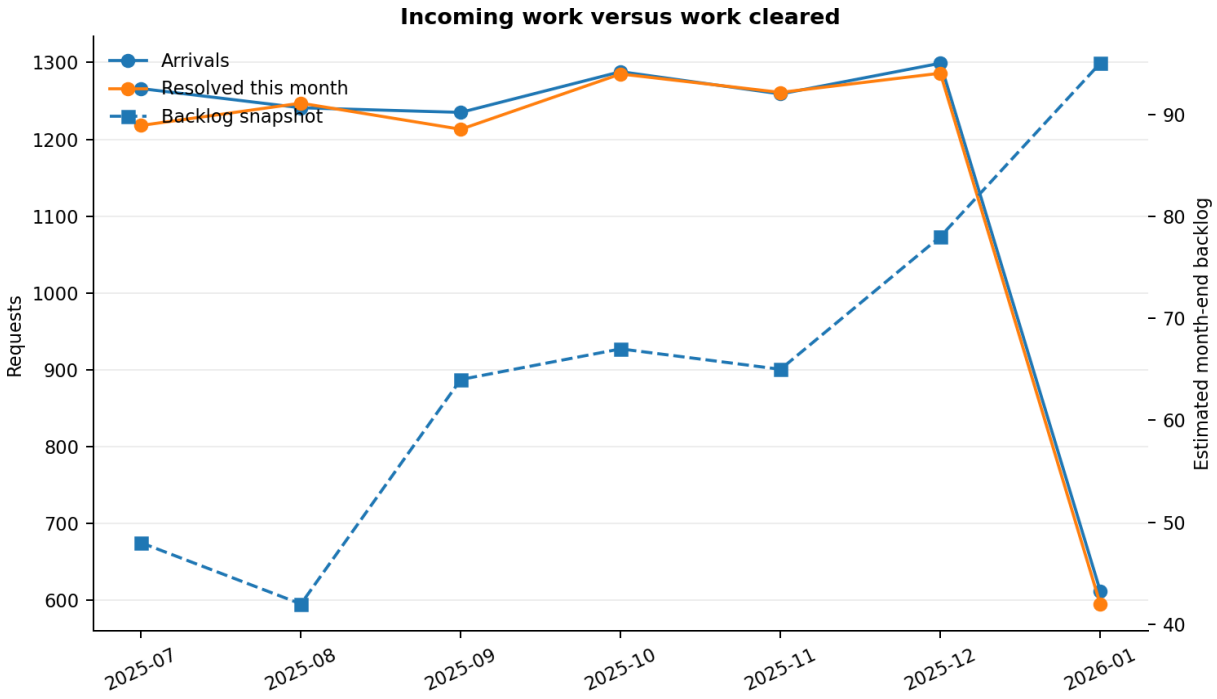


A median can look comfortable while a smaller group waits much longer. The report uses the 90th percentile alongside the median to avoid treating the majority experience as the whole story. The chart uses a log scale because request times range from short contacts to multi-day cases.

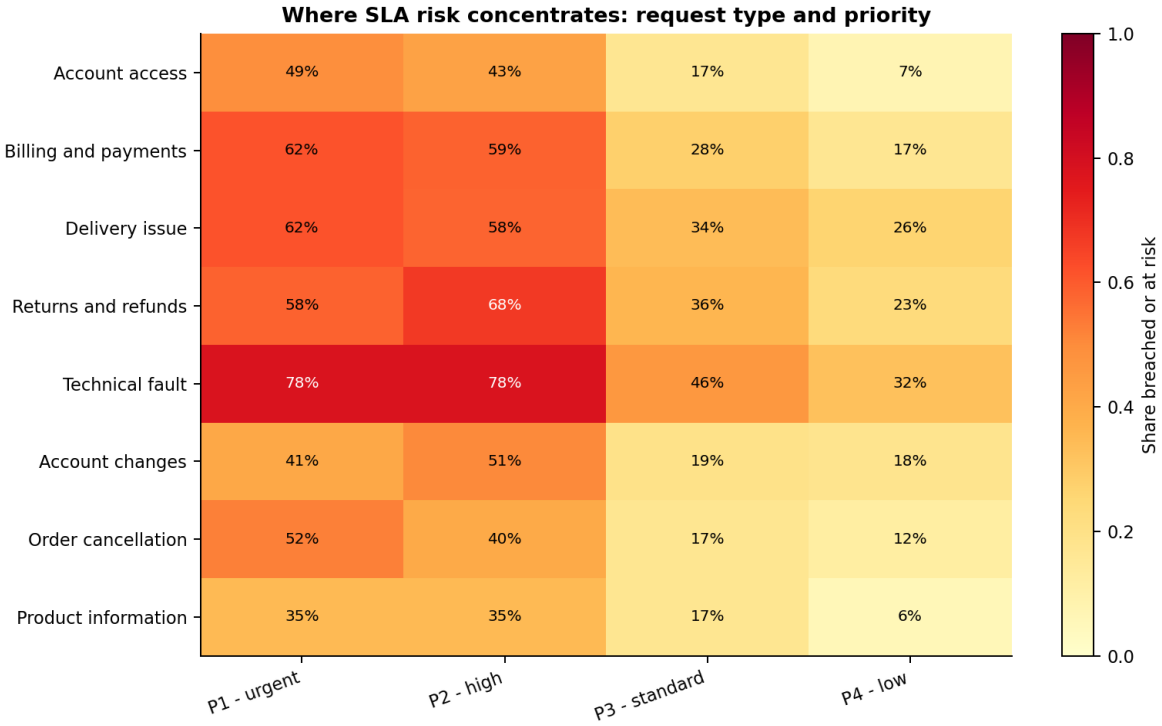


In the sample, the median among resolved cases with zero handoffs is 9.9 hours, compared with 13.8 hours for one handoff and 18.7 hours for two. This is an association, not proof that a handoff causes that entire delay: harder cases are more likely to be handed off. It is still a strong reason to inspect repeat handoffs and ownership changes on long-running requests.

Where pressure is building

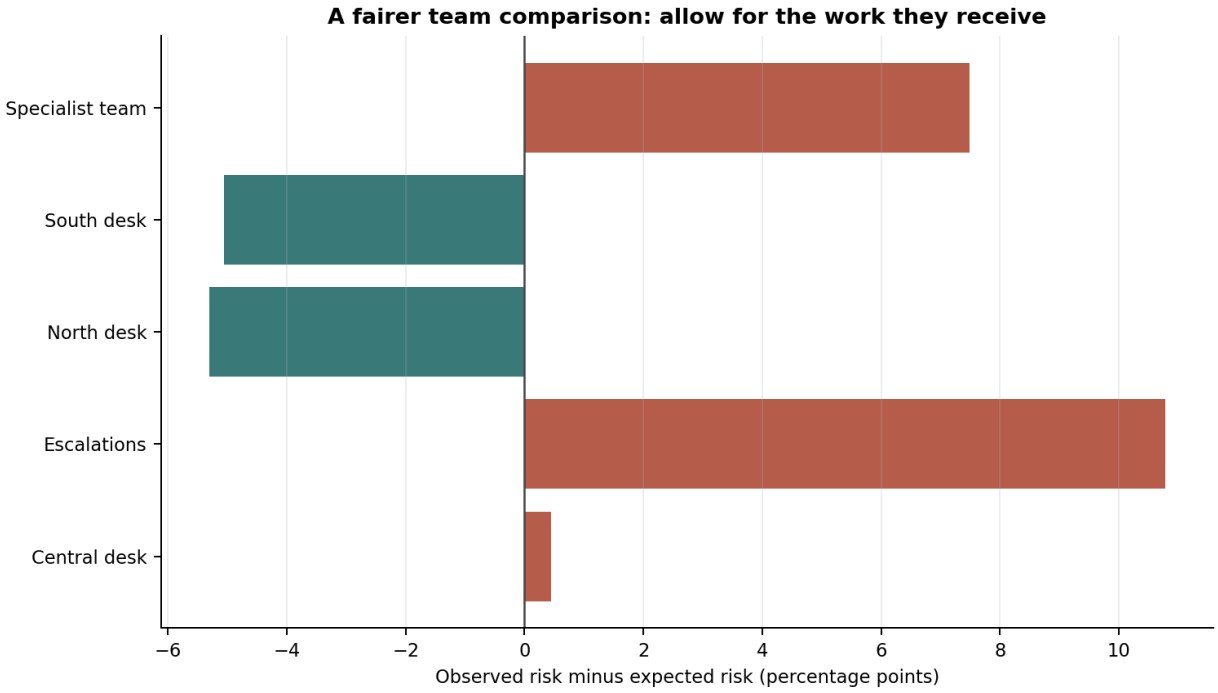


The monthly view compares incoming volume with work cleared and a reconstructed month-end backlog. The backlog measure is estimated from each request's arrival and elapsed time. It is useful for spotting pressure, but it is not a substitute for an actual daily queue snapshot or a capacity plan.



The heatmap helps separate service types where a stricter priority target may be doing its job from service types where work routinely outlasts its promise. Read low-volume cells cautiously; the same percentage can represent very different numbers of requests.

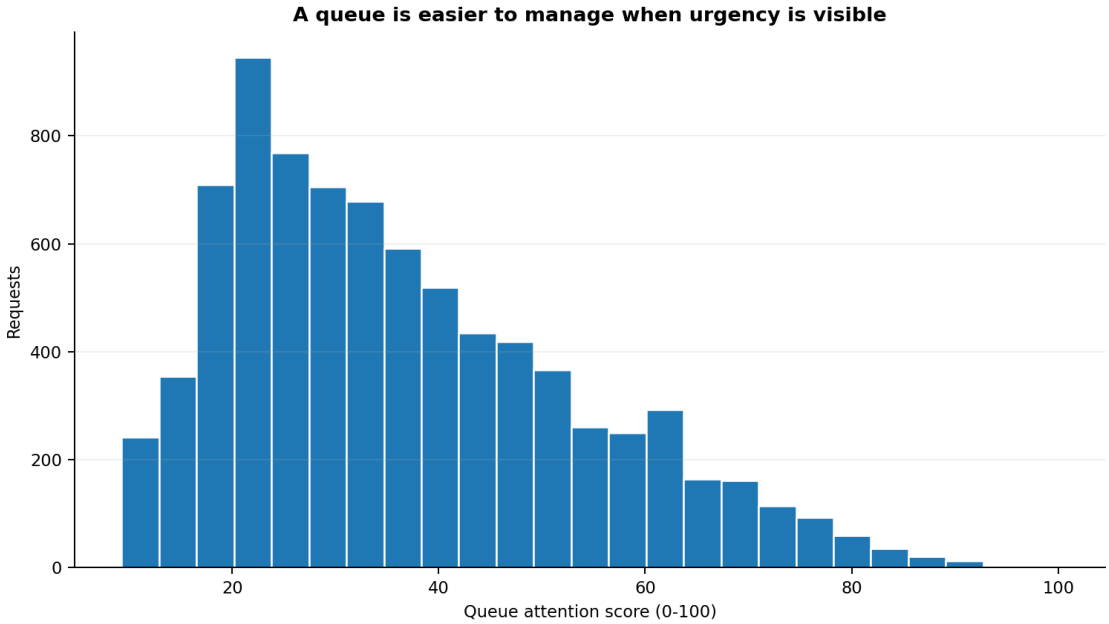
Compare teams without rewarding easy work



The team view adjusts each team's observed risk for the priority and service-type mix it received. North desk has the lowest risk gap in this synthetic sample, while Escalations has the highest. This is a lightweight case-mix adjustment, not a staff performance rating. It does not account for every factor, such as staffing levels, skill mix, shift coverage or live queue size.

The queue friction score

The highest-friction service type in this run is Technical fault, scoring 96 out of 100. The score blends percentile ranks for p90 elapsed time relative to target, average handoffs and the share breached or at risk. It is intentionally transparent: the score points managers toward queues to inspect; it does not replace operational judgement.



What to change first

1. Create an aged-work review, not just a backlog review. Start each shift with open requests that have consumed most or all of their target. Assign an owner and a next action, not only a new status.

2. Find the repeat handoff. Review the longest-running tickets with two or more handoffs. Look for missing intake information, unclear ownership, specialist queues that need a simpler decision tree, or transfers that could have been avoided.

3. Set queue-specific working rules. Start with the highest-friction service types and the priority cells with the highest late share. Use an explicit escalation threshold and make the threshold visible to the team.

4. Compare teams on like-for-like work. Keep the mix-adjusted view as a prompt for investigation. Add staffing, shift hours and incoming volume before using it to discuss team performance.

5. Run a small change before changing the whole operation. Pick one queue, reduce avoidable transfers or add a clear first-touch checklist, and compare its p90 time and SLA result against a similar queue over the same period. Record what else changed.

Method and cautions

All records in this repository are synthetic and were generated for portfolio demonstration. SLA targets are illustrative service rules attached to each priority, not an external standard. Open cases are counted as at risk when their elapsed age is greater than the target; resolved SLA performance is calculated separately. Team mix adjustment uses service type and priority strata only. The queue attention score is a triage aid, not a measure of customer value or staff quality. The generated activity file is a plausible event path rather than an audited history. No real customer, team or company data is included.

Reproduce the analysis with `python src/build_project.py` after installing requirements, or use the project's `run_analysis.py` entry point. Outputs include CSV scorecards, a SQLite database, the dashboard data payload and figures used in this brief.